

From Control Flow to Biological Evolution and Advancement

*Environmental Sensation, Awareness, Threat Discernment, Temporal Integrity, and
Presentation-State Integrity*

FOUNDATIONAL DESIGN BASIS - CHATGPT INDEPENDENT VERSION

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Program context	Omniversal Architect Labs (OAL) - Control Flow research; biological evolution/advancement; environmental sensing and threat discernment
Status	Draft for review - not canon - conceptual research design, not an operating protocol
Primary purpose	Preserve the origin, governing constraints, hypotheses, definitions, and first laboratory architecture for testing how entities discern environmental state and threats under uncertainty.
Safety posture	Simulation-first and benign-proxy-first. This document does not authorize exposure to poisons, toxins, corrosives, pathogens, or other hazardous agents.

Governing clarification from the Research Architect

Threat must not be minimized, euphemized, or hidden by the laboratory ontology. When evidence supports that a condition is a threat to the entity under test, the framework should name it as a threat and preserve its severity, urgency, uncertainty, and actionability. If the entity cannot discern the threat because its available sensory apparatus or information channels do not provide the necessary evidence, that limitation must be stated explicitly rather than converted into a finding of safety.

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1. Provenance and How the Topic Arose

This research thread begins in Control Flow and extends it into biological evolution and advancement, environmental sensation, awareness, threat detection, and state reconciliation. The originating problem is not simply how an organism senses. It is how an entity obtains enough trustworthy, time-correct, context-correct information to continue functioning when what is presented, what the environment is doing, and what the entity currently believes or experiences can diverge.

The originating questions ask how an organism knows what is around it; how that knowledge depends on temporal integrity and presentation-state integrity; how discrepancies among presentation, environment, and internal state are reconciled; how an entity distinguishes food from non-food, poison, toxin, corrosive exposure, predators, environmental hazards, and other dangers; how it constructs a reliable map from unreliable foundations; and how those capacities can be tested.

Two constraints materially changed the research problem. First, the laboratory does not assume a universal sensory apparatus or a human five-sense model. Second, threat cannot be treated as either universal or trivial. Not every entity is threatening and not every environment contains

active danger, but when an actual threat exists relative to an entity, the laboratory must not mask it behind neutral terminology in a way that reduces its operational significance.

This document is an independent ChatGPT design pass. It is informed by the user-originated concept, prior ChatGPT work on Presentation-State Integrity and Temporal Integrity / Temporal Aliasing, and the independently prepared Claude, Perplexity, and Grok foundational drafts supplied for comparison. It does not merge those drafts into consensus doctrine. Where this version agrees with them, the agreement is noted conceptually; where it differs, the independent design stance is retained.

Scope note on supplied attachments

One supplied Copilot attachment is a Tridactyl Manipulation Architecture document from a separate simulation-lab project. Because it addresses hand morphology and representation coherence rather than environmental threat discernment, it is treated as outside the evidence set for this foundational record.

2. Executive Determination

The laboratory should be built around a strong claim: an entity that inhabits an environment must be evaluated on whether it can obtain, reconcile, and act on the information that matters to its continued existence, integrity, function, goals, or other declared viability criteria. Threat discernment is therefore not a cosmetic label layered on top of environmental sensing. It is one of the highest-consequence outputs of environmental state estimation.

Neutral vocabulary is useful before classification, because the laboratory must not assume that every unknown condition is dangerous. However, neutrality ends when the evidence supports a threat relation. At that point, the system should preserve the classification rather than translate it into softer language such as anomaly, condition, variance, or environmental relation if those terms would conceal the possibility of harm. The laboratory must distinguish epistemic caution from semantic minimization.

The central scientific problem can be stated as: How can an entity construct and maintain an action-relevant model of its environment and itself from partial, noisy, delayed, contradictory, deceptive, inaccessible, or otherwise unreliable information, while preserving the ability to recognize and respond to threats when they are discernible and explicitly declaring when they are not?

Independent design stance

Threat is both relational and consequential. Relational means a condition becomes a threat with respect to a particular entity, state, pathway, time, and vulnerability. Consequential means that once the relation is supported, the design must not hide the danger behind abstractions. A threat can be context-dependent without being minimized.

3. Main Purpose and Scope

3.1 Main purpose

The purpose is to establish a testable, versioned foundation for studying environmental discernment under uncertainty, including threat discernment, resource recognition, map construction, state reconciliation, and adaptive action across heterogeneous sensory architectures.

3.2 The first laboratory domain

The first major experimental domain is a poison / toxin / corrosive discernment environment. These are useful stressors because they can vary by route, dose analogue, timing, detectability, delay, accumulation, entity structure, and presentation. They also expose a crucial distinction: a condition may be harmful without being directly sensible, and a detectable cue may be correlated with harm without being the causal agent.

3.3 In scope

- Biological, artificial, hybrid, robotic, simulated, distributed, or otherwise defined entities, provided the entity boundary and state variables are declared for each study.
- Open-ended sensory architectures including chemical, mechanical, thermal, electromagnetic, temporal, field-based, proprioceptive, interoceptive, social, instrument-mediated, synthetic, or currently hypothetical modalities.
- Temporal integrity, presentation-state integrity, state estimation, discrepancy detection, reconciliation, uncertainty, map construction, threat classification, and action selection.
- Known threats, partially known threats, latent threats, delayed threats, cumulative threats, ambiguous threats, threats outside the entity sensorium, and false threat cues.
- Simulation and benign physical testbeds that allow controlled manipulation without exposing people, animals, ecosystems, or living tissues to harmful agents.

3.4 Out of scope for this foundational version

- Operational chemical handling procedures, dosing instructions, exposure protocols, synthesis, procurement, or disposal instructions for real toxins, poisons, corrosives, pathogens, or hazardous agents.
- Claims that the framework describes all organisms, all environments, or all forms of threat.
- Claims that an entity must possess a human-like conscious representation in order to behave adaptively.
- Claims that laboratory telemetry is metaphysical ground truth. The laboratory maintains a reference model with known limitations.

4. Governing Design Laws

G1 - Sensory non-universality

Do not assume all entities use the same senses, the same number of senses, or human sensory categories. The test harness must be defined in terms of channels, transduction, accessibility, and discriminability

rather than a fixed five-sense template.

G2 - No optimal-evolution assumption

Do not assume an entity's sensory apparatus is perfectly evolved, optimally adapted, complete, reliable, or well matched to its current environment.

G3 - Threat non-universality

Do not assume every entity, object, event, process, or environment is threatening. Threat status is determined relative to a declared entity, context, time, pathway, protected variable, and evidence state.

G4 - Threat non-minimization

When the evidence supports a threat relation, name it and preserve its consequence. Do not use neutral vocabulary to make an actual threat appear less urgent, less harmful, or less relevant to continuation.

G5 - Discernibility declaration

For every tested threat, declare whether the threat is in principle discernible with the entity's available apparatus and information pathways. If it is not, record non-discernibility or insufficient evidence rather than failure-as-safety.

G6 - Non-detection is not safety

Absence of a detected threat means only that no threat was detected under the tested conditions. It does not prove the environment is safe.

G7 - Open-world threat boundary

The laboratory does not claim to discover, enumerate, or test all possible threats. Unknown and unmodeled threats remain possible.

G8 - Integrity separation

Temporal integrity, presentation-state integrity, sensing fidelity, causal attribution, internal-state estimation, and action capability are separate variables and should not be collapsed into one accuracy score.

G9 - Multiple explanations remain admissible

A discrepancy does not automatically imply deception, sensor failure, environmental change, or internal error. Competing explanations remain active until evidence discriminates among them.

G10 - Reference model fallibility

The laboratory's own environment register, sensors, clocks, labels, and scoring systems can be wrong. Reference-state integrity must itself be audited.

5. Relationship to Existing OAL Integrity Frameworks

5.1 Presentation-State Integrity

This laboratory inherits the independent ChatGPT Presentation-State Integrity Framework (PSIF) rather than redefining it from scratch. In that framework, Presentation-State Integrity is the degree to which an environment's claims, renderings, narratives, policies, interfaces, and explanations correspond to its evidenced operational state and causal structure. The key doctrine is that presentation is evidence about state, not state itself.

For this laboratory, PSI is applied to sensory and environmental presentation. A cue, percept, display, odor analogue, field measurement, internal signal, memory replay, social report, or instrumentation readout is a presentation available to the entity. The test asks whether that presentation preserves the state and causal relations needed for the entity to make a viable decision. A high-fidelity presentation may still be useless if the entity cannot access or interpret it; a compressed presentation may still be sufficient if it preserves the decision-relevant structure.

A mismatch should first be recorded as presentation-state divergence, not automatically as deception. Camouflage, mimicry, spoofing, omission, transformation, sensor saturation, stale data, instrumentation filtering, or benign projection effects can all create divergence. Intent is a separate evidentiary question.

5.2 Temporal Integrity and Temporal Aliasing

Temporal integrity concerns whether events, observations, internal states, actions, exposures, and outcomes remain correctly associated with acquisition time, order, duration, persistence, freshness, and causal window. This is broader than temporal aliasing.

The existing ChatGPT temporal framework reserves temporal aliasing for a specific sampling/reconstruction mechanism in which insufficient temporal sampling causes higher-frequency or differently ordered activity to appear as a different lower-frequency, reversed, or otherwise reconstructed identity. Clock drift, buffering, jitter, stale information, order inversion, duration distortion, temporal binding, postdiction, and attention-related temporal loss can all create temporal divergence without being strict aliasing.

That distinction matters for threat discernment. A threat may be perfectly detectable in one moment yet missed because its evidence is sampled too slowly, associated with the wrong

exposure, presented after the decision window closes, or fused with a different event. The laboratory should therefore record acquisition time, reception time, processing time, presentation time, action time, and outcome time where applicable.

5.3 Non-equivalence of presentation and reality

Core integrity doctrine

Underlying state -> partial observation -> transformation/sampling -> presentation -> interpretation -> reconciliation -> action. No stage is automatically identical to the stage before it. The laboratory evaluates the integrity of the transformations and the entity's capacity to recognize uncertainty or divergence.

6. Working Ontology and Definitions

Term	Working definition
Entity	The bounded system under test that can receive or otherwise acquire information, maintain state, and produce action, inaction, update, regulation, avoidance, refusal, or another measurable response.
Environment	The set of external and relational conditions that can affect the entity during the declared study interval. The laboratory models an environment; it does not claim exhaustive access to reality.
Protected variable / viability criterion	A declared quantity or condition whose unacceptable degradation constitutes harm for the trial: continuation, structural integrity, energy, control authority, memory integrity, mission state, reproductive analogue, map coherence, or another defined criterion.
Threat	A condition, process, event, object, organism, absence, delay, interaction, or information state that can produce an unacceptable adverse transition for the entity relative to a declared viability criterion, in a specified context and time window.
Hazard	A source or condition with the capacity to cause harm under some pathway or exposure. Hazard does not guarantee that the entity is currently threatened; threat requires relation to the entity and context.
Threat cue	An observable or inferable feature used by the

Term	Working definition
	entity to estimate a threat relation. The cue may be causal, correlated, misleading, absent, delayed, or inaccessible.
Discernibility	Whether sufficient information exists within the entity's accessible channels, temporal window, processing capacity, and action horizon to discriminate the threat-relevant state above a declared criterion.
Detectability	Whether a signal or state difference can be registered at all. Detection is weaker than identification or threat classification.
Identification	Assignment of a condition to a known or novel class with some confidence. Identification can be correct while threat appraisal is wrong, and vice versa.
Threat appraisal	Assessment that a detected or inferred condition may produce harm to the entity, including severity, imminence, probability/uncertainty, pathway, and available response.
Presentation state	The current rendered, surfaced, or operationally available representation of information to the entity. It may differ from source, observed, reconstructed, and interpreted states.
Entity state	The entity's current internal condition relevant to interpretation and action: damage, energy, memory, confidence, learned associations, goals, developmental state, sensor health, and control capacity.
Reference environmental state	The laboratory's best-supported, instrumented description of the tested environment for scoring. It is an operational reference, not an infallible metaphysical truth.
Reliable map	An internal or external representation whose confidence and structure are sufficiently calibrated to support the entity's available actions under tested conditions, including knowing when the map is stale, incomplete, ambiguous, or untrustworthy.

Term	Working definition
Unknown	A legitimate state in which evidence is insufficient for warranted classification. Unknown is not equivalent to safe, dangerous, or failure.

6.1 Threat statement schema

A threat claim should be operationalized rather than left as a free-floating label. Recommended form: Condition C is a threat to Entity E, in Context K, during Interval T, relative to Protected Variable V, through Pathway P, with Evidence/Uncertainty U, and with Action Set A available or unavailable.

This schema does not weaken the term threat. It makes the claim more precise and therefore harder to dismiss, dilute, or misapply. It also exposes cases in which a threat exists but the entity lacks a detection path, interpretation path, or viable response.

7. Independent Architecture

The architecture is functional rather than strictly serial. Biological and artificial systems may execute these functions in parallel, recursively, or through tightly coupled sense-act loops. The laboratory therefore scores functions and transitions rather than requiring a single cognitive pipeline.

Functional layer	Laboratory question
A. Environmental process	What exists or changes in the test world, including threats, resources, neutral conditions, latent variables, and unmodeled factors.
B. Presentation generation	How the environment becomes available as cues: emissions, surfaces, gradients, contact, social signals, telemetry, or laboratory displays.
C. Sensorium / access layer	What the entity can transduce, sample, receive, or otherwise access, including sensor limits, occlusion, saturation, damage, latency, and channel availability.
D. Temporal binding layer	How observations are time-indexed, ordered, grouped, kept fresh, and linked to exposures, actions, and consequences.
E. State estimation	How the entity estimates environmental state and its own internal state from available evidence.
F. Discrepancy detection	How the entity recognizes conflicts among channels, memory, prediction, internal state, and current presentation.

Functional layer	Laboratory question
G. Reconciliation / uncertainty	How competing hypotheses are preserved, weighted, tested, revised, or left unresolved.
H. Threat and resource appraisal	How the entity estimates harmful, beneficial, neutral, ambiguous, or unknown significance relative to its state and viability criteria.
I. Action selection / active sensing	How the entity acts, waits, samples, probes, avoids, approaches, seeks corroboration, or abstains. Action may be used to gather evidence, not only to respond.
J. Consequence and learning	How outcomes update maps, sensor trust, temporal associations, threat models, policies, and future confidence.
K. Laboratory audit layer	Independent records of reference state, timing, cue generation, entity-accessible information, and scoring; this layer is itself subject to integrity checks.

8. Hypotheses and Falsifiable Propositions

H1 - Discernment integration hypothesis

Viable threat discernment depends on the joint quality of sensory accessibility, temporal integrity, presentation-state integrity, internal-state estimation, discrepancy reconciliation, uncertainty calibration, and action availability, not on raw sensor count or detection accuracy alone.

H2 - Threat non-minimization hypothesis

When an actual threat is correctly identified but its representation is semantically downgraded or stripped of severity/action relevance, the probability of maladaptive action will increase relative to conditions in which threat status is preserved.

H3 - Discernibility-bound hypothesis

For threats whose evidence lies outside the entity's accessible sensory/information manifold, performance will not improve beyond chance or prior bias unless a new channel, tool, learned proxy, or active-sensing pathway becomes available. The correct laboratory interpretation is non-discernible under current conditions, not safe.

H4 - Integrity interaction hypothesis

Temporal-integrity failures and presentation-state failures will produce different error signatures and will interact nonlinearly when combined. A system robust to one may still fail under the other.

H5 - Calibrated-uncertainty hypothesis

Entities that can represent uncertainty and preserve multiple hypotheses will outperform equally sensitive entities forced into immediate binary safe/threat classification under ambiguous and partially observable conditions.

H6 - Active-sensing hypothesis

When uncertainty is reducible, entities capable of selecting information-gathering actions will improve threat discernment and reduce both false negatives and unnecessary false alarms compared with passive-only entities.

H7 - Map revision hypothesis

Reliable world models require not only accumulation of evidence but controlled forgetting, staleness detection, and revision. Entities that cannot downgrade outdated information will show systematic errors after environment changes.

H8 - Reference-layer integrity hypothesis

Apparent entity failure can be produced by corrupted laboratory reference state, clock disagreement, or mislabeled ground truth. Independent audit channels will reduce false attribution of failure to the entity.

9. Research Questions

1. What information about an environment is physically or computationally available to a given entity?
2. Which parts of that information can the entity actually transduce, discriminate, retain, and associate with time?
3. How does the entity distinguish sensor failure, environmental change, stale memory, ambiguous evidence, and internal-state change when all can produce similar discrepancies?
4. When a threat is present, is there a detectable cue? Is the cue accessible to the entity? Is it causally related to the threat or merely correlated?
5. What makes a threat discernible rather than merely detectable?
6. When the entity cannot discern a threat, can it recognize that its knowledge is insufficient?
7. How does the entity distinguish food, resource, neutral material, poison, toxin, corrosive condition, predator, structural hazard, information hazard, or other threat class without presuming human sensory categories?
8. How do dose analogue, route, duration, accumulation, latency, proximity, and entity state alter threat status?
9. How does temporal aliasing or other temporal divergence change apparent threat identity, persistence, direction, onset, or causal attribution?
10. How does presentation-state divergence caused by camouflage, mimicry, omission, filtering, or instrumentation change threat appraisal?
11. How does the entity decide which signals to trust when modalities disagree?

12. Can the entity use action as a probe to reduce uncertainty before committing to a high-cost decision?
13. How does it build a map that remains useful when the foundational observations are unreliable?
14. How are unknown unknowns represented in evaluation, given that the laboratory cannot enumerate all threats?
15. How do we tell the difference between correct behavior produced by correct understanding and correct behavior produced by coincidence, rigid reflex, or the wrong model?
16. What does the laboratory itself fail to observe, and how can reference-state uncertainty be quantified?

10. Laboratory Architecture and Staged Program

The foundational program should proceed in stages so that the epistemic machinery is tested before any high-consequence or biologically sensitive work is contemplated.

Stage	Purpose
Stage 0 - Formalization and governance	Declare entity class, environment, viability criteria, threat classes, sensorium, reference-state instrumentation, timing model, scoring rules, stopping conditions, and prohibited operations.
Stage 1 - Simulation	Use fully controlled simulated environments with explicit hidden state, synthetic cues, configurable sensor channels, adjustable latency/noise, and reversible threat proxies. This stage is the primary environment for testing ontology, metrics, temporal integrity, and presentation-state integrity.
Stage 2 - Benign physical testbed	Translate the simulation into nonhazardous physical proxies using lights, fields, temperature-safe ranges, inert materials, robotic obstacles, noninjurious chemical analogues, or instrumented surfaces. The goal is transfer, not danger.
Stage 3 - Nonliving / computational sensing platforms	Test heterogeneous sensors, robots, synthetic agents, or in vitro/computational models where scientifically justified. Preserve the same integrity and threat-discernibility accounting.
Stage 4 - Organism-level work, conditional	Only if scientifically necessary, ethically justified, legally permitted, and independently approved. The foundational document is not an authorization or protocol for this stage.

11. Threat Discernibility Framework

The laboratory must explicitly separate four questions that are often collapsed: Is a threat present? Is evidence of the threat available? Can the entity access that evidence? Can the entity interpret and act on it? Each question can have a different answer.

Threat-discernibility state	Interpretation
T0 - No supported threat	Reference evidence does not support a threat relation under the declared context. The entity should not be penalized for non-response.
T1 - Threat present and directly discernible	Threat-relevant evidence is available within the entity's accessible channels and decision window. Failure to detect or classify is a meaningful performance result.
T2 - Threat present, indirectly discernible	Direct threat signal is inaccessible, but reliable proxies, consequences, environmental traces, other entities, tools, or active-sensing actions can support inference.
T3 - Threat present but currently non-discernible	No sufficient accessible cue exists within the entity's apparatus, history, tools, or time window. The result should be marked non-discernible under current conditions, not safe and not necessarily an entity defect.
T4 - Threat present but evidence is temporally unavailable	Evidence exists but arrives too late, is stale, aliased, misordered, or outside the action window. This is a temporal-integrity limitation.
T5 - Threat present but presentation is misleading	Accessible cues systematically misrepresent the relevant state through camouflage, mimicry, omission, spoofing, transformation, or instrumentation artifacts. This is a presentation-state challenge.
T6 - Threat recognized but not addressable	The entity correctly discerns the threat but lacks a viable action, control surface, escape path, countermeasure, or sufficient time.
T7 - Threat uncertain / competing hypotheses	Evidence supports more than one plausible state. The correct behavior may be uncertainty preservation, active sensing, conservative action, or abstention depending on cost structure.
T8 - Unknown / unmodeled threat	The laboratory does not have a complete threat model. A later harmful outcome may reveal that the reference model itself omitted a threat

Threat-discernibility state	Interpretation
	class.

Non-minimization rule in scoring

T3, T4, T5, and T6 are not synonyms for low threat. They describe limitations in access, timing, presentation, or action. If the underlying condition is a high-consequence threat, the report should retain that consequence while separately describing why discernment or response failed.

12. Experiment Families

E1 - Sensorium profile and detection envelope

Measure what the entity can access across channel type, intensity, spatial/temporal scale, latency, saturation, interference, and sensor health. Output: an empirical detectability map, not an assumed sense inventory.

E2 - Threat vs non-threat discrimination

Present matched conditions where superficial cues are similar but threat relation differs. Measure sensitivity, criterion, false negatives, false positives, and confidence.

E3 - Poison / toxin / corrosive analogue suite

Use benign or simulated proxies representing intake-mediated harm, contact/material degradation, delayed consequences, cumulative exposure, and route-dependent effects. The entity must learn or infer which conditions threaten its declared viability criterion.

E4 - Cross-channel conflict

Provide conflicting cues across independently defined modalities. Test whether the entity fuses, segregates, discounts, or seeks more evidence, and whether it can detect common-mode corruption.

E5 - Temporal integrity

Manipulate delay, jitter, order, persistence, freshness, exposure duration, and consequence latency. Include strict temporal-aliasing cases separately from non-alias temporal errors.

E6 - Presentation-state integrity

Manipulate camouflage, mimicry, omission, partial truth, stale displays, mislabeled provenance, and high-fidelity but causally irrelevant cues.

E7 - Internal-state dependence

Hold environment constant while changing entity state variables such as energy, damage, memory, confidence, developmental stage, or learned policy. Determine whether threat meaning changes appropriately.

E8 - Active sensing

Allow the entity to perform probes that may reduce uncertainty. Score information gain, cost, delay, and whether the chosen probe actually discriminates competing hypotheses.

E9 - Map construction and remapping

Test navigation and threat/resource localization in environments that change after learning. Measure staleness detection, revision, overconfidence, and recovery.

E10 - Novelty and unknowns

Introduce conditions outside the training ontology. Measure whether the entity forces a known label, marks unknown, explores, freezes, or generalizes appropriately.

E11 - Threat outside the sensorium

Deliberately place a threat-relevant variable beyond the entity's accessible channels while keeping it visible to the audit layer. This tests whether the evaluation correctly distinguishes entity limitation from unsafe reality.

E12 - Reference-state corruption

Corrupt one laboratory clock, sensor, label, or ground-truth channel while preserving an independent oracle. Determine whether the scoring system can detect its own failure instead of blaming the entity.

E13 - Action-impossibility

Present correctly discernible threats for which no viable response exists. This tests the distinction between detection success and survival/control success.

E14 - Semantic threat minimization stress test

Keep the underlying threat and evidence constant but vary how downstream representations label it (explicit threat vs neutralized terminology). Test whether semantic compression or euphemism changes priority, action latency, or risk estimation.

13. Measures, Scorecards, and Interpretation

Measure	Meaning
Detection sensitivity	Ability to register threat-relevant signal differences independent of decision threshold.
Decision criterion	Bias toward calling ambiguous conditions threatening or safe; useful for separating caution from improved sensitivity.
False-negative rate	Threat-present trials classified as non-threat/safe or ignored. High-consequence misses should remain visible even if overall accuracy is high.

Measure	Meaning
False-positive rate	Non-threat trials classified as threats. Must be interpreted relative to the cost of caution.
Time-to-discernment	Elapsed time from availability of sufficient evidence to a justified threat appraisal.
Action latency	Elapsed time from justified appraisal to selected response.
Confidence calibration	Whether reported/implicit confidence tracks actual reliability.
Unknown calibration	Whether the entity uses unknown/insufficient-evidence states when warranted rather than forcing classification.
Temporal integrity score	Correct association of evidence with time, order, freshness, duration, and causal window.
Presentation-state integrity score	Degree to which the entity's available presentation preserves decision-relevant relations to the reference state and source/provenance.
Discrepancy detection rate	Ability to recognize conflicts among channels, internal model, memory, and current presentation.
Reconciliation quality	Ability to preserve alternatives, seek discriminating evidence, update appropriately, and avoid premature closure.
Map calibration	Relationship between map confidence and actual action utility across stable and changed environments.
Threat salience retention	Whether confirmed threat status, severity, and urgency survive transformations into internal or external representations.
Capability-bound attribution	Whether failures are correctly assigned to sensing, timing, presentation, inference, action, reference state, or irreducible non-discernibility rather than collapsed into entity failure.
Continuation / viability outcome	Whether the entity preserves the declared viability criterion, reported separately from detection and interpretation.

No single composite score should replace the underlying measures during early research. A system can survive by luck, detect a threat too late, correctly classify but fail to act, or act correctly for the wrong reason. The scorecard must preserve those distinctions.

14. Failure Modes and Red-Team Conditions

Failure mode	Why it matters
Anthropocentric sensor leakage	The environment is designed so that the only useful cues happen to align with human vision/hearing/smell/taste/touch categories.
Threat dilution	Actual danger is relabeled as neutral variance or anomaly in a way that reduces action priority.
Threat inflation	Unknown or merely unusual conditions are automatically labeled dangerous.
Non-detection equals safety	The evaluation treats absence of entity response as evidence that no threat exists.
Reference oracle overconfidence	Laboratory telemetry is assumed infallible despite clock, sensor, labeling, or implementation failure.
Temporal collapse	Acquisition, receipt, processing, presentation, and action times are reduced to one timestamp.
Alias overreach	Every timing discrepancy is called temporal aliasing, obscuring the actual mechanism.
Presentation capture	What is rendered or reported is treated as complete reality without mechanism or provenance testing.
Causal overclaim	A cue correlated with harm is treated as causal without intervention or discriminating evidence.
Outcome-only scoring	Survival or task success is treated as proof of correct perception and reasoning.
Binary-forcing	The entity is forced to choose safe/threat when the evidence supports unknown, ambiguous, or non-discernible.
Action conflation	Failure to neutralize a threat is scored as failure to detect it, even when no viable response exists.
Unknown-threat blindness	The lab assumes its threat catalog is exhaustive

Failure mode	Why it matters
	and therefore misclassifies unmodeled harm as entity error or noise.
Semantic masking by abstraction	A framework term becomes so generic that it stops carrying the actual consequence experienced by the entity.

15. Alternative Analysis and Dissent Preservation

Alternative analysis should be a built-in method rather than an appendix added only when disagreement occurs. This is especially important because the project explicitly compares independent AI-authored frameworks and because the laboratory itself studies disagreement among representations.

Alternative analysis	Use
Competing ontology analysis	Run the same scenario under more than one operational threat definition and report where classifications diverge.
Sensor-ablation analysis	Remove or degrade one channel at a time to identify which information actually supports discernment.
Cost-structure swap	Change the consequences of false negatives and false positives without changing sensory evidence, revealing whether the entity adapts decision criterion.
Reference-state audit	Use at least one independent channel or preserved event record to test the lab's own labels and timing.
Counterfactual entity analysis	Hold environment constant while changing sensorium, memory, learning, or action capabilities.
Threat-salience audit	Trace a confirmed threat through every transformation layer and verify that severity, urgency, uncertainty, and response requirements are not silently stripped away.
Dissent log	Record disagreements among analysts/models as explicit unresolved alternatives rather than forcing premature consensus.
Negative-space register	Record important variables that were not sensed, not measured, not modeled, or not testable. Absence from the dataset must remain

Alternative analysis	Use
	visible.
Disconfirmation-first review	For each favored explanation, specify what evidence would falsify it and actively search for that evidence.

16. Safety, Ethics, and Governance Boundary

This foundational record is a conceptual research design, not an operating protocol. The poison/toxin/corrosive domain should begin with simulation and benign proxies. Nothing in this document authorizes procurement, preparation, dosing, release, exposure, handling, or disposal of hazardous substances or biological agents.

- Use simulation or harmless surrogates for early validation of sensing, timing, mapping, and threat-classification logic.
- Separate conceptual ownership from safety and regulatory authority. Any future work involving people, animals, living tissue, hazardous chemicals, or ecosystems requires qualified institutional review and domain-specific controls.
- Do not expose an entity to avoidable harm merely to prove that it can detect harm. Prefer reversible, noninjurious, information-equivalent tests whenever possible.
- Preserve stop conditions and fail-safe mechanisms at a layer independent of the entity under test when the study could create consequential states.
- Never infer that a simulated success validates real-world toxicology or biological safety.

17. What Must Be Resolved Before Implementation

17. What entity class is the first target: simulated organism, robotic agent, artificial control system, or another bounded entity?
18. What is the first declared viability criterion, and what counts as unacceptable change?
19. Which threat classes belong in Series 1, and which are intentionally deferred?
20. Which sensory channels are available to the entity, and which are available only to the audit layer?
21. What makes a tested threat directly discernible, indirectly discernible, or non-discernible?
22. What is the independent reference-state architecture, and how will its own integrity be tested?
23. Which time bases are used, and how are acquisition, transmission, processing, presentation, action, and consequence timestamps preserved?
24. What uncertainty representation is available to the entity?
25. What active-sensing actions are permitted, and what do they cost?
26. What are the pass/fail or advancement gates for each experimental family?
27. How are false negatives weighted when the consequence is entity termination or irreversible loss?
28. How will semantic threat minimization be detected in reports, interfaces, or internal representations?

29. How will unknown threats and unmodeled conditions be logged without pretending they were tested?
30. Which portions of the prior PSI and Temporal Integrity frameworks are imported as canonical dependencies versus merely referenced?
31. What governance mechanism decides when an independent draft becomes OAL canon?

18. Recommended First Simulation Campaign

The recommended first campaign should not begin with real toxic or corrosive agents. It should create a simulation in which the laboratory knows the full programmed state but the entity receives only a controlled subset. The purpose is to establish the discernibility accounting, threat-salience rules, temporal integrity, and presentation-state integrity before expanding scope.

Scenario	Purpose
S1-A Baseline	No threat. Multiple neutral stimuli. Establish false-positive rate and ordinary map construction.
S1-B Direct threat cue	A simulated threat has an accessible, reliable cue. Measure detection, classification, confidence, and action.
S1-C Inaccessible threat	Threat exists only in audit state. Entity has no relevant channel. Expected result: non-discernible, not safe.
S1-D Proxy-only threat	Direct threat variable is hidden, but a reliable secondary cue is accessible. Test inference and learning.
S1-E Temporal delay	Threat cue is correct but stale or delayed beyond the optimal response window. Separate sensing success from temporal-integrity failure.
S1-F Presentation mismatch	Threat cue is transformed, camouflaged, or contradicted by another channel. Test discrepancy detection and active sensing.
S1-G State-dependent threat	Same environmental condition is safe in one entity state and threatening in another. Test relational classification.
S1-H No-action threat	Threat is correctly discerned but no viable action exists. Confirm that scoring does not convert action failure into detection failure.
S1-I Semantic minimization	Threat is correctly identified, but downstream labels vary in explicitness. Test whether neutralized labels reduce response priority.

Scenario	Purpose
S1-J Reference corruption	One audit channel is wrong while a second oracle is preserved. Test whether the evaluation can identify laboratory-side integrity failure.

Advancement criterion: the laboratory should not progress to more realistic or higher-consequence environments until it can reliably distinguish threat presence, threat discernibility, entity detection, entity interpretation, temporal failure, presentation failure, action limitation, and reference-state failure as separate outcomes.

19. Versioning and Provenance Discipline

- Every revision retains the document ID, increments the version, records date and preparer, and lists substantive changes.
- Architect-originated constraints remain distinguishable from model-generated hypotheses and interpretations.
- Imported definitions from PSIF and Temporal Integrity remain traceable to their source documents rather than being silently rewritten.
- Companion AI drafts remain independent records. Agreement does not automatically create canon; disagreement is preserved in the dissent log.
- Experimental findings must be separated from foundational assumptions. A successful simulation result does not retroactively turn a design premise into a biological fact.
- Terminology changes that affect threat, integrity, discernibility, or scoring require corresponding updates to measures and traceability tables.

Appendix A. Threat-State Matrix

Reference threat state	Presentation/cue state	Entity access	Interpretation
No threat	No cue	None needed	Correct non-response is possible; do not force threat behavior.
No threat	Threat-like cue	Accessible	False-positive challenge; measure caution and criterion.
Threat	Reliable direct cue	Accessible	Direct discernment test.
Threat	Reliable indirect cue	Accessible	Inference / learned-proxy test.
Threat	Cue exists	Inaccessible	Non-discernible under current sensorium.
Threat	Cue available too late	Accessible after window	Temporal-integrity limitation.
Threat	Misleading cue	Accessible	Presentation-state integrity challenge.
Threat	Correctly recognized	Accessible	No viable action; separate recognition from control.
Threat uncertain	Conflicting cues	Accessible	Uncertainty calibration and active sensing.
Unmodeled harm	Unknown to lab ontology	Varies	Reference-model limitation; update ontology without rewriting prior evidence.

Appendix B. Minimum Event Record

Field	Purpose
Trial / event ID	Stable cross-reference
Entity ID / version	Which entity architecture was tested
Environment version	Which world/configuration generated the event

Field	Purpose
Viability criterion	What state is protected
Threat state	Reference classification and uncertainty
Threat pathway	How harm could occur
Discernibility class	T0-T8 or successor taxonomy
Sensorium configuration	Available channels and limits
Acquisition time	When a channel sampled/received evidence
Processing / presentation time	When evidence became usable to the entity
Action time	When response occurred
Outcome time	When consequence became observable
Presentation provenance	Source and transformations applied
Entity internal state	Relevant energy/damage/memory/confidence/context
Entity classification	Threat/non-threat/unknown/etc.
Confidence	Declared or inferred confidence
Action / inaction	What the entity did
Outcome	Effect on viability criterion
Laboratory reference channels	Evidence supporting score
Integrity flags	Temporal, presentation, reference, or other anomalies
Alternative explanations	Competing interpretations retained
Reviewer disposition	Confirmed, contradicted, unresolved, invalid trial

Appendix C. Source and Companion-Document Register

Record	Role in this document
User-originated research statements (21 Sep 2026)	Controlling source for intent: Control Flow -> biological evolution/advancement; environmental sensing; state reconciliation; poison/toxin/corrosive laboratory; sensory non-universality; open-world threat constraints; threat non-minimization clarification.

Record	Role in this document
Presentation_State_Integrity_Framework_Independent_Development_v1.0 - ChatGPT	Predecessor internal framework. Supplies PSI definition, state-claim divergence discipline, presentation-is-evidence doctrine, provenance, correction, and separation of mismatch from intent.
OAL-TI-001 ChatGPT Temporal Integrity / Aliasing / Reconciled Presentation series	Predecessor internal temporal framework. Supplies distinction between Temporal Integrity as umbrella and strict temporal aliasing as one mechanism, plus temporal provenance and sampling/reconstruction concerns.
foundational-document-environmental-sensing - Claude independent draft	Companion independent analysis. Contributes state-estimation, signal-detection, active-inference, identifiability-limit, and threat-ontology perspectives.
Threat_Discernment_Foundation_v0.1 - Perplexity independent draft	Companion independent analysis. Contributes detailed threat relation schema, integrity dimensions, staged program, measures, and explicit decision gates.
OAL-CF-EST-FOUND-Grok-v0.1.0 - Grok independent draft	Companion independent analysis. Contributes entity-integrity criterion, register separation, laboratory-reference fallibility, negative-space thinking, and alternative-analysis structures.
OAL-SL-TDA-COP-1.0 Tridactyl Simulation Lab - Copilot	Supplied attachment but outside this topic. It concerns tridactyl manipulation architecture and was not used as a substantive source for this threat-discernment foundation.

Appendix D. Change Log

Version	Date	Preparer	Change
v0.1.0	21 Sep 2026	ChatGPT - GPT-5.6 Sol	Initial independent foundational version. Incorporated Research Architect feedback that threat must not be minimized; imported prior ChatGPT PSI and Temporal Integrity definitions; established threat-

Version	Date	Preparer	Change
			discernibility classes, architecture, hypotheses, simulation campaign, measures, alternative-analysis methods, and safety boundary.

Foundational conclusion

The laboratory is not complete when it can label an environment. It is complete only when it can explain what the entity could sense, what it could not sense, what was presented, what was temporally available, what state the entity was in, why a threat was or was not discernible, whether the threat was accurately represented rather than minimized, what actions were available, and what remains unknown. A threat that cannot be discerned is still a threat if the evidence supports that relation; the laboratory must preserve both truths at once: the danger exists, and the entity may lack the means to know it.